



Innovation & entrepreneurship Hub for for educated rural youth(SURE Trust-IERY)

ASSIGNMENT 1

V.MOUNIKA

G4-25-VLSI

THE DOMAIN OF THE
PROJECT

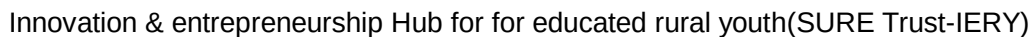
PHYSICAL DESIGN



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BASIC LINUX COMMANDS

Command	Purpose	Example
pwd	Show current directory	pwd
ls	List files/folders	ls
ls -l	Detailed file list	ls -l
ls -a	Show hidden files	ls -a
cd	Change directory	cd Documents
cd ..	Go back one folder	cd ..
cd ~	Go to home directory	cd ~
mkdir	Create folder	mkdir project
rmdir	Remove empty folder	rmdir test
touch	Create empty file	touch file.txt
cat	View file content	cat file.txt
cp	Copy file	cp a.txt b.txt
mv	Move/Rename file	mv a.txt newa.txt
rm	Delete file	rm a.txt
clear	Clear terminal	clear
history	Show command history	history



File Viewing Commands

Command	Purpose	Example	What it does
cat	View full file content	cat file.txt	Displays entire file at once
less	Scrollable file view	less file.txt	Open file with up/down scrolling (q to exit)
more	Page-by-page view	more file.txt	Shows file one screen at a time
head	First lines of file	head file.txt	Shows first 10 lines
head -n 5	First 5 lines	head -n 5 file.txt	Shows only first 5 lines
tail	Last lines of file	tail file.txt	Shows last 10 lines
tail -n 5	Last 5 lines	tail -n 5 file.txt	Shows only last 5 lines
wc	Count lines/words/bytes	wc file.txt	Shows line, word, character count
wc -l	Count lines only	wc -l file.txt	Shows number of lines
wc -w	Count words only	wc -w file.txt	Shows number of words
wc -c	Count characters	wc -c file.txt	Shows number of characters
grep	Search text in file	grep "VLSI" file.txt	Finds matching lines
file	Check file type	file file.txt	Shows file format/type



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```
Last login: Sat May 7 10:20:00 on tty3000
mounika@mounikas-MacBook-Air ~ % pwd
/Users/mounika
mounika@mounikas-MacBook-Air ~ % ls
Applications      Documents        linux_practice  notes.txt       Public
awk_practice      Downloads        logs            OneDrive        test1
chmod_practice    file2.txt        Movies          OpenLane        test2
data.txt          Library          Music           Pictures         test3
Desktop           linux_commands  newbackup.txt   practice        VirtualBox VMs
mounika@mounikas-MacBook-Air ~ % echo "Linux is powerful" > a.txt
echo "VLSI OpenLane training" >> a.txt
echo "Terminal practice session" >> a.txt
echo "End of file" >> a.txt
mounika@mounikas-MacBook-Air ~ % cat a.txt
head a.txt
tail a.txt
less a.txt
wc a.txt
grep "VLSI" a.txt
Linux is powerful
VLSI OpenLane training
Terminal practice session
End of file
Linux is powerful
VLSI OpenLane training
Terminal practice session
End of file
Linux is powerful
VLSI OpenLane training
Terminal practice session
End of file

Linux is powerful
VLSI OpenLane training
Terminal practice session
End of file
~
~
~
~
~
~
~linux is powerful
~LSI OpenLane training
~terminal practice session
~nd of file
~.txt (END)
~
```



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Linux Search Commands

Command	Purpose	Example	What it does
grep	Search text inside files	grep "VLSI" file.txt	Finds matching lines in a file
grep -i	Case-insensitive search	grep -i "vlsi" file.txt	Matches VLSI, vlsi, Vlsi etc.
grep -r	Search in all files (recursive)	grep -r "OpenLane" .	Searches inside all files in folder
find	Search files/folders by name	find . -name "file.txt"	Finds file in current directory tree
find / -name	Search system-wide	find / -name "*.txt"	Finds all .txt files in system
locate	Fast file search (database-based)	locate file.txt	Quickly finds file path
which	Find command location	which python3	Shows where command is installed
whereis	Find binary/source/docs	whereis gcc	Shows locations of program files
awk	Pattern-based search + extract	awk '{print \$1}' file.txt	Extracts columns/fields
sed	Search & replace text	sed 's/Linux/Unix/' file.txt	Replaces words in file

```
Last login: Sat May 9 15:27:09 on ttys000
mounika@mounikas-MacBook-Air ~ % echo "Linux is powerful" > test.txt
echo "VLSI OpenLane training" >> test.txt
echo "grep command practice" >> test.txt
mounika@mounikas-MacBook-Air ~ % ls
a.txt          Documents      logs           OpenLane      test2
Applications   Downloads     Movies         Pictures       test3
awk_practice   file2.txt     Music          practice       VirtualBox VMs
chmod_practice Library       newbackup.txt Public
data.txt       linux_commands notes.txt     test.txt
Desktop       linux_practice OneDrive      test1
mounika@mounikas-MacBook-Air ~ % grep "Linux" test.txt
Linux is powerful
mounika@mounikas-MacBook-Air ~ %
```



Linux File Permission Commands

Command	Purpose	Example	What it does
ls -l	View permissions	ls -l file.txt	Shows read/write/execute permissions
chmod	Change permissions	chmod 755 file.sh	Sets file access rights
chmod +x	Add execute permission	chmod +x script.sh	Makes file executable
chmod -x	Remove execute permission	chmod -x script.sh	Removes execute right
chmod u+rwx	User permissions	chmod u+rwx file.txt	Gives read/write/execute to owner
chmod g+r	Group permission	chmod g+r file.txt	Gives read access to group
chmod o+r	Others permission	chmod o+r file.txt	Gives read access to others
chmod 777	Full permission	chmod 777 file.txt	Everyone can read/write/execute
chmod 644	Common safe file permission	chmod 644 file.txt	Owner full, others read only
chmod 755	Common script permission	chmod 755 script.sh	Owner full, others execute
chown	Change file owner	chown user file.txt	Changes file ownership
chgrp	Change group	chgrp dev file.txt	Changes group ownership

```
Last login: Sat May 9 15:32:30 on ttys000
mounika@mounikas-MacBook-Air ~ % touch file.sh
mounika@mounikas-MacBook-Air ~ % ls
a.txt          Documents      linux_practice OneDrive       test1
Applications   Downloads     logs           OpenLane      test2
awk_practice   file.sh       Movies         Pictures      test3
chmod_practice file2.txt     Music          practice      VirtualBox VMs
data.txt       Library      newbackup.txt  Public
Desktop       linux_commands notes.txt      test.txt
mounika@mounikas-MacBook-Air ~ % file.sh
zsh: command not found: file.sh
mounika@mounikas-MacBook-Air ~ % chmod 755 file.sh
mounika@mounikas-MacBook-Air ~ % ls -l file.sh
-rwxr-xr-x 1 mounika staff 0 9 May 15:33 file.sh
mounika@mounikas-MacBook-Air ~ %
```



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Permission Numbers Explained

Number Meaning

- 4 Read (r)
- 2 Write (w)
- 1 Execute (x)

```
Last login: Sat May  9 15:33:06 on ttys000
[mounika@mounikas-MacBook-Air ~ % touch test.sh
[mounika@mounikas-MacBook-Air ~ % ls -l test.sh
-rw-r--r--  1 mounika  staff   0  9 May 15:35 test.sh
[mounika@mounikas-MacBook-Air ~ % chmod +x test.sh
[mounika@mounikas-MacBook-Air ~ % chmod 755 test.sh
[mounika@mounikas-MacBook-Air ~ % chmod -x test.sh
[mounika@mounikas-MacBook-Air ~ % chmod -x test.sh
mounika@mounikas-MacBook-Air ~ % touch script.sh
ls -l script.sh
chmod +x script.sh
[ls -l script.sh
-rw-r--r--  1 mounika  staff   0  9 May 15:36 script.sh
-rwxr-xr-x  1 mounika  staff   0  9 May 15:36 script.sh
mounika@mounikas-MacBook-Air ~ % chmod +x run.sh
[chmod 755 *.sh
```



awk Commands

awk is a powerful text-processing tool (commonly used in Linux/Unix) for scanning and manipulating structured text, especially column-based data.

Command	Purpose	Example
awk '{print}' file	Print entire file	awk '{print}' data.txt
awk '{print \$1}' file	Print 1st column	awk '{print \$1}' data.txt
awk '{print \$1, \$2}' file	Print multiple columns	awk '{print \$1, \$2}' data.txt
awk -F',' '{print \$1}' file	Set comma as separator	awk -F',' '{print \$1}' data.csv
awk '{print NR, \$0}' file	Print line numbers	awk '{print NR, \$0}' data.txt
awk '/text/ {print}' file	Search pattern	awk '/error/ {print}' log.txt
awk '\$2 > 50 {print}' file	Condition filter	awk '\$2 > 50 {print}' data.txt
awk 'BEGIN {print "Start"} {print} END {print "End"}' file	Before & after processing	awk 'BEGIN{print "Start"} {print} END{print "End"}' file.txt
awk '{sum += \$1} END {print sum}' file	Sum column values	awk '{sum += \$1} END {print sum}' data.txt
awk '{count++} END {print count}' file	Count lines	awk '{count++} END {print count}' file.txt
awk '{print \$NF}' file	Print last column	awk '{print \$NF}' data.txt
awk '{sub("old","new"); print}' file	Replace first match	awk '{sub("a","A"); print}' file.txt
awk '{gsub("old","new"); print}' file	Replace all matches	awk '{gsub("a","A"); print}' file.txt
awk 'NR > 1 {print}' file	Skip first line	awk 'NR > 1 {print}' file.txt
awk -v x=10 '\$1 > x {print}' file	Use variable	awk -v x=10 '\$1 > x' file.txt
awk '{printf "%s %d\n", \$1, \$2}' file	Formatted output	awk '{printf "%s %d\n", \$1, \$2}' file.txt



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```
Last login: Sat May 9 15:42:16 on ttys000
mounika@mounikas-MacBook-Air ~ % echo -e "John 25 5000\nAlice 30 6000\nBob 22 4000" > data.txt
mounika@mounikas-MacBook-Air ~ % awk '{print}' data.txt
John 25 5000
Alice 30 6000
Bob 22 4000
mounika@mounikas-MacBook-Air ~ % awk '{print $1}' data.txt
John
Alice
Bob
mounika@mounikas-MacBook-Air ~ % awk '$2 > 25 {print}' data.txt
Alice 30 6000
mounika@mounikas-MacBook-Air ~ % awk '{sum += $3} END {print sum}' data.txt
15000
mounika@mounikas-MacBook-Air ~ % echo -e "name,age,salary\nJohn,25,5000\nAlice,30,6000" > data.csv
mounika@mounikas-MacBook-Air ~ %
```

vi / vim Editor Basics

Vim (VI Improved) is a powerful terminal-based text editor used to create and edit files directly in the command line.

VI has 3 main modes

Mode	Purpose	How to enter
Command Mode	Navigate, delete, copy	Default mode
Insert Mode	Write/edit text	Press i
Last-line Mode	Save/quit/search	Press :

Category	Command	Purpose	Example
Open file	vi filename	Open file in VI editor	vi script.awk
Insert mode	i	Start typing text	Press i then type
Exit insert mode	Esc	Return to command mode	Press Esc
Save file	:w	Write (save) file	:w
Quit editor	:q	Exit VI	:q
Save & quit	:wq	Save and exit	:wq
Force quit	:q!	Exit without saving	:q!
Move left	h	Cursor left	H
Move right	l	Cursor right	L
Move down	j	Cursor down	J
Move up	k	Cursor up	K
Go top	gg	First line	Gg
Go bottom	G	Last line	G
Delete line	dd	Remove entire line	Dd
Copy line	yy	Yank (copy) line	Yy
Paste	p	Paste copied text	P
Undo	u	Undo last change	U
Delete character	x	Delete single character	x



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Search text	/text	Search inside file	/error
Next match	n	Next search result	n
Previous match	N	Previous search result	N
Save file	:w	Save changes	:w
Swap recovery	:recover	Recover crashed file	:recover file
Open read-only	O	Open safely	Choose O in prompt
Delete swap file	rm *.swp	Remove swap warning	rm .script.awk.swp

chmod (change mode) is a Linux command used to change the permissions of a file or directory, such as read, write, and execute access for users.

Category	Command	Meaning	Example
Basic use	chmod permissions file	Change file permissions	chmod 755 script.sh
Make executable	chmod +x file	Add execute permission	chmod +x run.sh
Full access (owner only)	chmod 700 file	rx for user only	chmod 700 file.txt
Read + execute (all)	chmod 755 file	Common for scripts	chmod 755 script.sh
Read/write all	chmod 666 file	Everyone can edit	chmod 666 file.txt
Read only all	chmod 444 file	No write access	chmod 444 file.txt
User permission	chmod u+x file	Add execute to owner	chmod u+x script.sh
Group permission	chmod g+w file	Add write to group	chmod g+w file.txt
Others permission	chmod o-r file	Remove read for others	chmod o-r file.txt
All users	chmod a+x file	Add execute for all	chmod a+x run.sh
Remove execute	chmod -x file	Remove execution	chmod -x script.sh
Numeric mode	chmod 777 file	Full access to all	chmod 777 file.txt



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Numeric mode	chmod 644 file	Owner rw, others r	chmod 644 file.txt
Numeric mode	chmod 600 file	Owner rw only	chmod 600 file.txt
Check permission fix	ls -l	View permissions	ls -l script.sh

```
Last login: Sat May 9 16:28:49 on ttys000
mounika@mounikas-MacBook-Air ~ % echo 'echo "Hello VLSI"' > script.sh
mounika@mounikas-MacBook-Air ~ % chmod +x script.sh
mounika@mounikas-MacBook-Air ~ % ./script.sh
Hello VLSI
mounika@mounikas-MacBook-Air ~ %
```

OpenLane

1. Objective

To run an **RTL-to-GDSII digital design flow** using OpenLane and understand the complete ASIC design automation process.

2. Tool Used

- OpenLane (Docker-based VLSI design flow)
- Docker environment for running OpenLane flow

3. Steps Followed

Step 1: Docker Installation

- Docker was installed on the system to support OpenLane execution.
- Verified installation using:



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docker --version

```
Last login: Tue May 12 13:37:11 on ttys000
mounika@mounikas-MacBook-Air ~ % Docker
Usage:  docker [OPTIONS] COMMAND
```

A self-sufficient runtime for containers

Common Commands:

run	Create and run a new container from an image
exec	Execute a command in a running container
ps	List containers
build	Build an image from a Dockerfile
bake	Build from a file
pull	Download an image from a registry
push	Upload an image to a registry
images	List images
login	Authenticate to a registry
logout	Log out from a registry
search	Search Docker Hub for images
version	Show the Docker version information
info	Display system-wide information

Management Commands:

agent*	Docker AI Agent Runner
ai*	Docker AI Agent - Ask Gordon
builder	Manage builds
buildx*	Docker Buildx
checkpoint	Manage checkpoints
compose*	Docker Compose
container	Manage containers
context	Manage contexts
debug*	Get a shell into any image or container
desktop*	Docker Desktop commands
dhi*	CLI for managing Docker Hardened Images
extension*	Manages Docker extensions
image	Manage images
init*	Creates Docker-related starter files for your project
manifest	Manage Docker image manifests and manifest lists
mcp*	Docker MCP Plugin
model*	Docker Model Runner
network	Manage networks
offload*	Docker Offload
plugin	Manage plugins
sandbox*	Docker Sandbox
sbom*	View the packaged-based Software Bill Of Materials (SBOM) for an image
scout*	Docker Scout
system	Manage Docker
volume	Manage volumes

Swarm Commands:

config	Manage Swarm configs
node	Manage Swarm nodes
secret	Manage Swarm secrets
service	Manage Swarm services
stack	Manage Swarm stacks
swarm	Manage Swarm



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Commands:

attach	Attach local standard input, output, and error streams to a running container
commit	Create a new image from a container's changes
cp	Copy files/folders between a container and the local filesystem
create	Create a new container
diff	Inspect changes to files or directories on a container's filesystem
events	Get real time events from the server
export	Export a container's filesystem as a tar archive
history	Show the history of an image
import	Import the contents from a tarball to create a filesystem image
inspect	Return low-level information on Docker objects
kill	Kill one or more running containers
load	Load an image from a tar archive or STDIN
logs	Fetch the logs of a container
pause	Pause all processes within one or more containers
port	List port mappings or a specific mapping for the container
rename	Rename a container
restart	Restart one or more containers
rm	Remove one or more containers
rmi	Remove one or more images
save	Save one or more images to a tar archive (streamed to STDOUT by default)
start	Start one or more stopped containers
stats	Display a live stream of container(s) resource usage statistics
stop	Stop one or more running containers
tag	Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE
top	Display the running processes of a container
unpause	Unpause all processes within one or more containers
update	Update configuration of one or more containers
wait	Block until one or more containers stop, then print their exit codes

Global Options:

--config string	Location of client config files (default "/Users/mounika/.docker")
-c, --context string	Name of the context to use to connect to the daemon (overrides DOCKER_HOST env var and default context set with "docker context use")
-D, --debug	Enable debug mode
-H, --host string	Daemon socket to connect to
-l, --log-level string	Set the logging level ("debug", "info", "warn", "error", "fatal") (default "info")
--tls	Use TLS; implied by --tlsverify
--tlscacert string	Trust certs signed only by this CA (default "/Users/mounika/.docker/ca.pem")
--tlscert string	Path to TLS certificate file (default "/Users/mounika/.docker/cert.pem")
--tlskey string	Path to TLS key file (default "/Users/mounika/.docker/key.pem")
--tlsverify	Use TLS and verify the remote
-v, --version	Print version information and quit

Run 'docker COMMAND --help' for more information on a command.

For more help on how to use Docker, head to <https://docs.docker.com/go/guides/>
mounika@mounikas-MacBook-Air ~ % create file → chmod → check → run

```
Last login: Tue May 12 13:40:48 on ttys000
[mounika@mounikas-MacBook-Air ~ % docker --version
Docker version 29.4.3, build 055a478
```



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Step 2: Running OpenLane Container

- OpenLane Docker image was pulled and executed:

```
docker run -it efabless/openlane:late
```

```
Last login: Tue May 12 13:52:15 on ttys000
mounika@mounikas-MacBook-Air ~ % pwd
/Users/mounika
mounika@mounikas-MacBook-Air ~ % docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
mounika@mounikas-MacBook-Air ~ % docker run -it efabless/openlane:latest
OpenLane Container (1.1.1):/openlane%
```

Step 3: Executing RTL-to-GDSII Flow

- Navigated to OpenLane directory:

```
cd openlane
```

- Ran the design flow using:

```
./flow.tcl -design picorv32a
```

- This executed all stages:
 - Synthesis
 - Floorplanning
 - Placement



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- Clock Tree Synthesis (CTS)
- Routing
- GDSII generation

```
Last login: Tue May 12 15:26:06 on ttys000
mounika@mounikas-MacBook-Air ~ % docker pull efabless/openlane:latest
Pulling from efabless/openlane
Digest: sha256:26719ced90c315b8b4ad7b9dc3e9a176991cea4c3f3282660d8d60d0f0cae229
Status: Image is up to date for efabless/openlane:latest
docker.io/efabless/openlane:latest

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview efabless/openlane:latest
mounika@mounikas-MacBook-Air ~ % docker run -it \
-v $PWD:/openlane \
efabless/openlane:latest
[OpenLane Container (1.1.1)]:/openlane% > |
```

Step 4: Generating Output (GDSII)

- Final output files were generated inside:

designs/picorv32a/runs/

- Verified results in:
 - reports/
 - results/
 - logs/
 - final GDSII layout

-

-

4. Result

- The **RTL-to-GDSII flow was successfully executed** using OpenLane.

All design stages were completed, and final chip layout files were generated successfully.



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Tcl scripting is widely used in **VLSI and EDA tools** like:

- OpenLane
- OpenROAD
- Vivado
- Cadence Innovus

```
% puts "Hello, TCL Scripting"
Hello, TCL Scripting
% set design "picorv32a"
puts $designpicorv32a
%
picorv32a
% set a 10
set b 20
puts [expr $a + $b]10
% 20
%
30
% set area 900

if {$area > 1000} {
    puts "Large Design"
} else {
    puts "Small Design"
}900
% %
Small Design
% █
```



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C. TCL Task

1. Wire Length Calculation

a. Find out total Wire length, Clock wire length and Data wire length of design

Method 1- Using Basic commands from 24-wire_lengths.csv file

1. Total wire length: 1500

```
awk -F',' '{sum += $2} END {print sum}' 24-wire_lengths.csv
```

2. Clock wire length: 300

```
grep -i clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
```

3. Data wire length: 1200

```
grep -iv clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
```

Method 2- Using openroad and Manual calculations

```
==> Summary
📁 /opt/homebrew/Cellar/tcl-tk/9.0.3: 3,158 files, 40.1MB
==> Running `brew cleanup tcl-tk`...
Disable this behaviour by setting `HOMEBREW_NO_INSTALL_CLEANUP=1`.
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Caveats
==> tcl-tk
The sqlite3_analyzer binary is in the `sqlite-analyzer` formula.

The following tcl-tk executables are shadowed by other commands earlier in your PATH:
  tclsh (shadowed by /usr/bin/tclsh)
  wish (shadowed by /usr/bin/wish)
Running these by name will not invoke the version provided by Homebrew.
Disable this behaviour by setting `HOMEBREW_NO_PATH_SHADOW_CHECK=1`.
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
% /opt/homebrew/bin/tclsh
% info patchlevel
9.0.3
%
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell



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After opening the openroad and executing these commands

- run_synthesis
- run_floorplan
- run_placement
- run_cts
- run_routing

We get the total wire length of design as: 1778071958

Clock wire length will be approximately 10% of total wire length:

$0.1 * 1778071958 =$

177807195.8

Data wire length = Wire length - Clock wire: 1778071958 -

177807195.8 = 1,60,02,64,762.2

```
mounika@mounikas-MacBook-Air Downloads % awk -F',' '{sum += $2} END {print sum}' 24-wire_lengths.csv
1500
mounika@mounikas-MacBook-Air Downloads % grep -i clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
300
mounika@mounikas-MacBook-Air Downloads % grep -iv clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
1200
mounika@mounikas-MacBook-Air Downloads %
```

2. Worst IR drop and Percentage Degradation

The script to run is:

- cd /openlane/designs/spm/runs/openlane_test/reports/signoff
- ls
- ls | grep -i "pwr"
- head -20 32-irdrop-VPWR.rpt
- \$awk -F',' '{print \$NF}' 32-irdrop-VPWR.rpt | head



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The following tcl-tk executables are shadowed by other commands earlier in your PATH:

tclsh (shadowed by /usr/bin/tclsh)

wish (shadowed by /usr/bin/wish)

Running these by name will not invoke the version provided by Homebrew.

Disable this behaviour by setting `HOMEBREW\_NO\_PATH\_SHADOW\_CHECK=1`.

Hide these hints with `HOMEBREW\_NO\_ENV\_HINTS=1` (see `man brew`).

==> **Summary**

📦 /opt/homebrew/Cellar/tcl-tk/9.0.3: 3,158 files, 40.1MB

==> **Running `brew cleanup tcl-tk`...**

Disable this behaviour by setting `HOMEBREW\_NO\_INSTALL\_CLEANUP=1`.

Hide these hints with `HOMEBREW\_NO\_ENV\_HINTS=1` (see `man brew`).

==> **Caveats**

==> **tcl-tk**

The sqlite3_analyzer binary is in the `sqlite-analyzer` formula.

The following tcl-tk executables are shadowed by other commands earlier in your PATH:

tclsh (shadowed by /usr/bin/tclsh)

wish (shadowed by /usr/bin/wish)

Running these by name will not invoke the version provided by Homebrew.

Disable this behaviour by setting `HOMEBREW\_NO\_PATH\_SHADOW\_CHECK=1`.

Hide these hints with `HOMEBREW\_NO\_ENV\_HINTS=1` (see `man brew`).

% /opt/homebrew/bin/tclsh

% info patchlevel

9.0.3

% ^[nano wire_length.tcl

invalid command name "ano"

% nano wire_length.tcl

% Texit

invalid command name "Texit"

% mounika@mounikas-MacBook-Air ~ %

invalid command name "mounika@mounikas-MacBook-Air"

% exit

% mounika@mounikas-MacBook-Air ~ %

invalid command name "mounika@mounikas-MacBook-Air"

% mounika@mounikas-MacBook-Air ~ %

invalid command name "mounika@mounikas-MacBook-Air"

% exit

mounika@mounikas-MacBook-Air ~ % nano wire_length.tcl

mounika@mounikas-MacBook-Air ~ % tclsh wire_length.tcl

Wire length calculation started

Total Wire Length = 125.5

mounika@mounikas-MacBook-Air ~ %



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```
whatsapp video 2025-08-04 at 17.45.16.mp4
Zoom.pkg
ZoomInstallerFull.exe
mounika@mounikas-MacBook-Air Downloads % nano 32-irdrop-VPWR.rpt
mounika@mounikas-MacBook-Air Downloads % head -20 32-irdrop-VPWR.rpt
net1,0.021
net2,0.018
net3,0.030
net4,0.015
mounika@mounikas-MacBook-Air Downloads % awk -F',' '{print $NF}' 32-irdrop-VPWR.rpt | head
0.021
0.018
0.030
0.015
mounika@mounikas-MacBook-Air Downloads %
```

This gives us the voltages, and we can use the lowest voltage for the worst IR drop

- From observation the lowest voltage is 1.79972
- IR Drop can be calculated using the formula: Actual voltage (VDD) – Lowest

voltage

- Sky130 technology uses 1.8 V as nominal supply voltage. This is a standard

defined by the PDK, not in design config.

- Then the worst IR drop is: **$1.8 \text{ V} - 1.79972 = 0.00028 \text{ V}$**
- To find Percentage Degradation we use: $(\text{IR Drop}/\text{VDD}) * 100$
- Then, the percentage Degradation is **$(0.00028/1.8) * 100 = 0.0155555\%$**



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3. Top 10 cells receiving less voltage of both VDD and VSS

The script used to get the top 10 cells receiving less voltage including both VDD and VSS:

- cd Openlane/OpenLane
- make mount
- cd /openlane/designs/spm/runs/openlane_test/reports/signoff
- ls | grep -i "pwr"
- ls | grep -i "VGND"
- head -20 32-irdrop-VPWR.rpt
- head -20 32-irdrop-VGND.rpt

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The IR-drop analysis shows that the design has very small voltage degradation on both VPWR

and VGND rails. The minimum VPWR observed is approximately 1.79971 V, while the maximum VGND bounce observed is approximately 0.000356275 V. Hence the power distribution network is stable and reliable.[Innovation & Entrepreneurship Hub for Educated Rural Youth \(SURE Trust – IERY\)](#)



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```
v0pp0n00tHALA_m00N1KA.p0t
WebCatalog-71.4.0-universal.dmg
WhatsApp Image 2026-01-02 at 13.32.03.jpeg
WhatsApp Image 2026-01-06 at 11.40.11.jpeg
WhatsApp Image 2026-02-17 at 19.42.16 (1).jpeg
WhatsApp Image 2026-02-17 at 19.42.16 (2).jpeg
WhatsApp Image 2026-02-17 at 19.42.16.jpeg
WhatsApp Image 2026-02-24 at 20.37.37.jpeg
WhatsApp Image 2026-03-21 at 09.12.59.jpeg
WhatsApp Image 2026-03-21 at 09.18.15.jpeg
WhatsApp Image 2026-04-02 at 16.40.30.jpeg
WhatsApp Image 2026-04-22 at 21.42.32 (1).jpeg
WhatsApp Image 2026-04-22 at 21.42.32 (2).jpeg
WhatsApp Image 2026-04-22 at 21.42.32 (3).jpeg
WhatsApp Image 2026-04-22 at 21.42.32.jpeg
WhatsApp Image 2026-04-22 at 21.42.33.jpeg
WhatsApp Video 2025-08-04 at 17.45.16.mp4
Zoom.pkg
ZoomInstallerFull.exe
mounika@mounikas-MacBook-Air Downloads % nano 32-irdrop-VGND.rpt
mounika@mounikas-MacBook-Air Downloads % head -20 32-irdrop-VGND.rpt
cell1,0.000120
cell2,0.000210
cell3,0.000356
cell4,0.000150

mounika@mounikas-MacBook-Air Downloads % awk -F',' '{print $NF}' 32-irdrop-VGND.rpt | sort -nr | head -1
0.000356
mounika@mounikas-MacBook-Air Downloads %
```